Phenomena: The Giant Pumpkin Mystery

Focus question: Where does a pumpkin's mass come from?

I notice…	I wonder…

Part 1: My initial model

Draw where you think the pumpkin's mass comes from.

Source	Soil	Water	Air	Sunlight	Fertilizer	The seed
My rank (before data)						
My rank (after data)						

Phenomena: The Giant Pumpkin Mystery (continued)

Part 2: What are these things made of?

Element in the pumpkin	Possible source(s)
Carbon (C)	
Hydrogen (H)	
Oxygen (O)	
Nitrogen (N)	

4.

5.

Part 3: Our questions

Exit ticket

What Are Living Things Made Of?

Focus question: Which elements make up the molecules of life, and where could a plant get them?

Warm-up

Part 1: The sealed-bag demo

Step	Mass (g)
Sealed bag, before tipping the cup	
My prediction after tipping (up, down, or same?)	
Sealed bag, after tipping the cup	
After opening the bag	

Part 2: Reading a formula

Formula	C	H	O	N	Other
CO_2 (carbon dioxide)					
H_2O (water)					
$C_6H_{12}O_6$ (glucose)					

What Are Living Things Made Of? (continued)

Part 3: Who needs the soil?

Watch: Plants Need More Than C, H, and O (3 min). Then fill in the quick sort with a partner.

Molecule	Formula	Type of big molecule	CO_2 + H_2O only? (Y/N)	Extra element(s) from soil
Glucose	$C_6H_{12}O_6$			
Oleic acid	$C_{18}H_{34}O_2$			
Valine	$C_5H_{11}NO_2$			
AMP	$C_{10}H_{14}N_5O_7P$			
Challenge: chlorophyll a	$C_{55}H_{72}MgN_4O_5$			

Atom accounting rule

A plant can build _____ from only CO_2 and water. To build _____, it also needs _____ from the _____.

Exit ticket

Building Sugar from Air and Water

Focus question: Where do the atoms in sugar come from, and what does light do?

Warm-up

Part 1: A pumpkin leaf in a sealed chamber

Part 2: Build sugar with beads

	Black (C)	White (H)	Red (O)	Total beads
Before (6 CO_2 + 6 H_2O)				
After (1 glucose + 6 O_2)				

Building Sugar from Air and Water (continued)

Part 3: Energy or matter?

Item	M or E	Item	M or E
Sunlight		Oxygen gas	
Carbon dioxide		Heat given off by a leaf	
Water		Nitrogen in fertilizer	
Glucose		Chlorophyll	

Exit ticket

Vocabulary and SketchNotes

Focus question: What words and big ideas will we need to solve the pumpkin mystery?

Frayer chart: 10 words

Word	Definition (my words)	Facts	Example (draw)	Non-example
Atom				
Element				
Molecule				
Conservation of matter				
Photosynthesis				
Glucose				
Amino acid				
Protein				
Monomer				
Polymer				

Vocabulary and SketchNotes (continued)

SketchNotes: Sugar to Structures

The 5 big ideas: a short phrase + a simple icon for each. One color for matter, another for energy. No full sentences.

Sensemaking: the Pumpkin Path

Draw the pumpkin plant with air, water, soil, and sunlight. Solid arrows = matter, dashed arrows = energy. Add one idea from each of Days 1–4, then keep adding after each lesson.

From Sugar to Amino Acids

Focus question: How can a plant turn sugar into amino acids?

Warm-up

Part 1: The amino acid recipe

Part 2: Build amino acids

From Sugar to Amino Acids (continued)

Part 3: Make sense of the model

What's the Evidence?

Focus question: What evidence do scientists have for how plants build their molecules?

Home group #: _____ My station #: _____

Part 1: Expert notes (my station)

Question	My answer
1	
2	
3	
4	
5 (Station 3 only)	

The most important thing my station shows:

What's the Evidence? (continued)

Part 2: Home group evidence table

Station	What they did	Key data (numbers)	What it shows	Claim(s)
1. Labeled carbon				
2. Tomatoes, no N				
3. Squash + fungi				
4. Extra CO_2				

Exit ticket: CER (Station 2)

Part	My answer
Claim	
Evidence (2 numbers)	
Reasoning	

Unit CER evidence log (back of notebook): add today's row.

Small Pieces, Big Molecules

Focus question: How do amino acids become proteins, and what happens to them when an animal eats?

Warm-up

Part 1: Monomers and polymers

Monomer (small)	Polymer (large)	Job in a pumpkin
Amino acid		
Glucose	Starch	
Glucose		Tough cell walls
Nucleotide		

Starch and cellulose contain: _____ Proteins contain: _____

Part 2A: Build a pumpkin seed protein

Amino acid	V	L	T	K	Q
How many?					

Small Pieces, Big Molecules (continued)

Part 2B: Eat, digest, and rebuild

Amino acid	V	L	T	K	Q
Hair protein needs					
Short or extra (+/−)					

Same protein as the pumpkin's? ____ Where did its atoms come from? ____

Part 3: Trace a carbon atom

Step	Where is the carbon atom?	What happens to it?
1	In a ____ molecule in the air	A pumpkin leaf takes it in.
2	In a ____ molecule in the leaf	Built during ____ using ____ energy
3	In an ____	Combined with ____ from the soil
4	In a pumpkin seed ____	Linked in a specific ____
5	In an amino acid in your blood	You ate the seed and ____ it.
6	In your hair protein	Your cells ____ them in a new order.

Exit ticket

Solving the Pumpkin Mystery

Focus question: What is the pumpkin made of, and how did it get built?

Warm-up

Part 1: Class consensus model

Draw: what goes into the leaf · what the leaf builds · what comes from the soil · what the plant builds from sugar + soil elements. Label each arrow with its evidence.

Part 2: Revise my Day 1 model

Then I thought…	Now I think…	Because (evidence)…

Solving the Pumpkin Mystery (continued)

Part 3: Explain it to the grower (Unit CER)

Checklist	My explanation
Claim: what the pumpkin is made of	
Evidence: 2 pieces of grower data	
Evidence: 1 piece from class	
Reasoning: sugar from CO_2 + water, then + N	
Correct the grower: soil and plant food	
Energy vs. matter: sunlight	

Part 4: Peer scoring

Scored by	Score (1–4)	Glow	Grow

Review: Sugar to Structures

Focus question: Which targets do I still need to work on before the CSA?

Self-rating (1–4)

Target	Do Now	After stations
LS1-6.1 Elements in sugar and large molecules		
LS1-6.2 Taking in and rearranging matter		
LS1-6.3 Sugar atoms plus other elements		
LS1-6.4 Trace atoms with a model		
LS1-6.5 Explain with evidence		
LS1-6.6 Revise an explanation		

Review stations

Station	One thing I learned or fixed
A · Atoms and elements	
B · Sugar plus nitrogen	
C · Trace the atoms	
D · Evidence and revision	

Review: Sugar to Structures (continued)

Driving question

How does a plant build its body, and what is it built from? Use at least 4 vocabulary words.

My assigned CFA reviews

Reviews open for me: _____ Done (check off): _____

CSA Reflection

Focus question: How do atoms from sugar become the structures of living things?

After the CSA: rate yourself (1–4)

Target	Day 9	Today
LS1-6.1 Elements in sugar and large molecules		
LS1-6.2 Taking in and rearranging matter		
LS1-6.3 Sugar atoms plus other elements		
LS1-6.4 Trace atoms with a model		
LS1-6.5 Explain with evidence		
LS1-6.6 Revise an explanation		

Which target improved the most since Day 1? Why?